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College of Science and Technology
Rinchending: Bhutan

CSF101
Programming Methodology
(SS2026)

Practical 1 Report

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Date: 13/03/2026

RUB Wheel of Academic Law: Academic Dishonesty

Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorised group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication**: the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration**: making a false declaration to receive special consideration

by an Examination Board or to obtain extensions to deadlines or exemption from work.

5. **Falsification of data**: presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism**: the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

Aim:

Set up a programming development environment on Windows **by installing** Python**,** Visual Studio Code**,**
the Python (Visual Studio Code extension)**, and** Git**.**

Theory:

Python: it is one of the famous programming languages used by many others users all over the world. It has simple and easily readable syntax and good language for beginner who are going through programming on the first time. Python is used in versatile field like in web development, data analysis, artificial intelligence, and automation. This is due to its big number of libraries and community, enhancing programmer to create application more faster and efficient.

VS Code:

It is an easy and efficient code editor created by Microsoft. users typically use it in order to write and maintain their code supporting most programming languages like Python, Java, and C++. It offers a handy functionality such as syntax highlighting, debugger tools, extensions, and code suggestions, which simplifies and increases coding efficiency to the developers.

GitHub:

It is web-based service, enabling the programmer to store, manage and share their programming code with others. It is founded on the Git version control system that assists in monitoring the changes in a project with time. GitHub is utilized in group work by developers to collaborate, and contribute to the open-source software. Additionally it maintains different forms of a project and enhance team work in software development to a great extent.

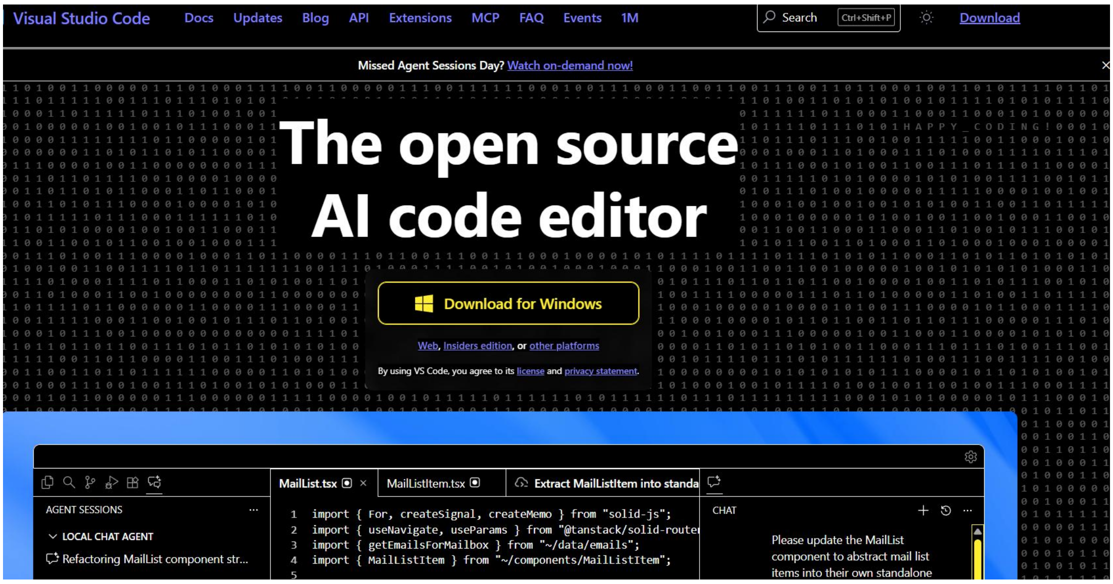
Implementation Steps

Step-by-step procedure to install Python on Windows: (pre-installed)

1. Step-by-step procedure to install Python on Windows:
Step 1: Go to the Python website
Open your browser and go to the official Python website:
python.org
Step 2: Download Python
On the homepage, click **Downloads**.
You will usually see a big button like:
Download Python 3.x.x
Step 3: Open the installer
After the file finishes downloading, open the installer.
Step 4: Important checkbox
Before clicking install, make sure to check:
Add Python to PATH
This is very important, because it lets you run Python from Command Prompt.
Step 5: Install Python
Click:
Install Now
Wait for the installation to complete.**Step 6: Finish installation**
Once it says setup was successful, click **Close**.
Step 7: Verify installation
Open **Command Prompt** and type:
python --version
If Python is installed properly, it will show something like:
Python 3.x.x
You can also check pip by typing:
pip --version
Step 8: Test Python
In Command Prompt, type:
python
Then try:
print("Hello, world!")
If it works, Python is installed correctly.
To exit Python, type:
exit()
Alternative if python does not work
Sometimes on Windows, you may need to use:

py --version
or
py
instead of python.

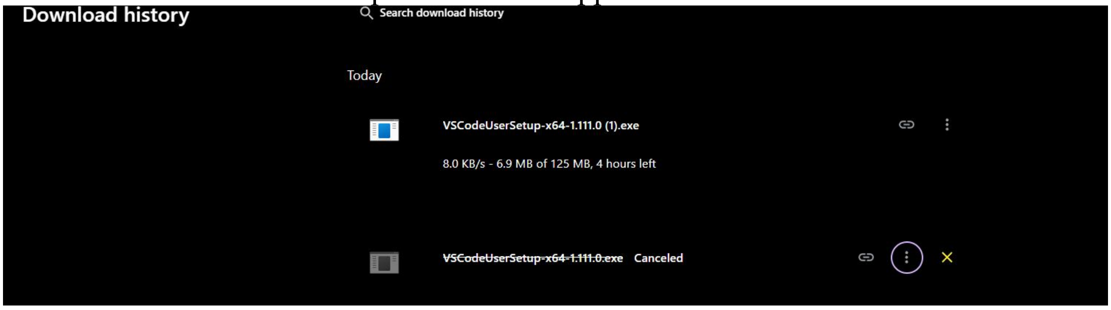
2. Step-by-step procedure to install Visual Studio Code (VS Code) on Windows
Step 1: Open the Visual Studio Code website
Open your web browser and go to the official website:
https://code.visualstudio.com



Step 2: Download VS Code
On the homepage, click the button:
Download for Windows
The installer file will start downloading.

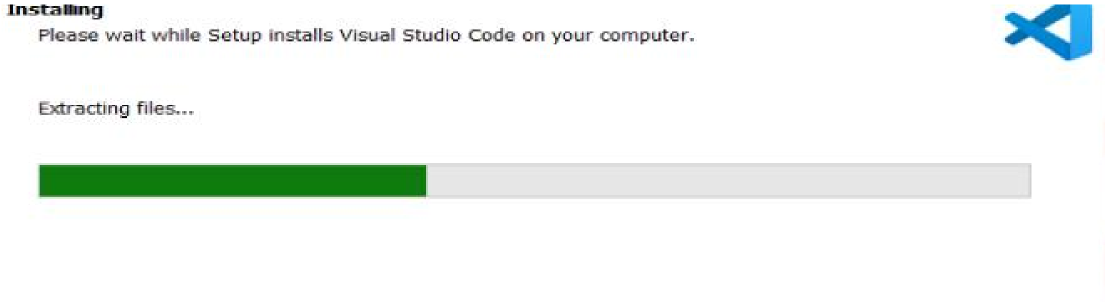
Step 3: Run the installer

After the download is complete, open the downloaded file:
VSCodeUserSetup-x64.exe
The Visual Studio Code setup window will appear.



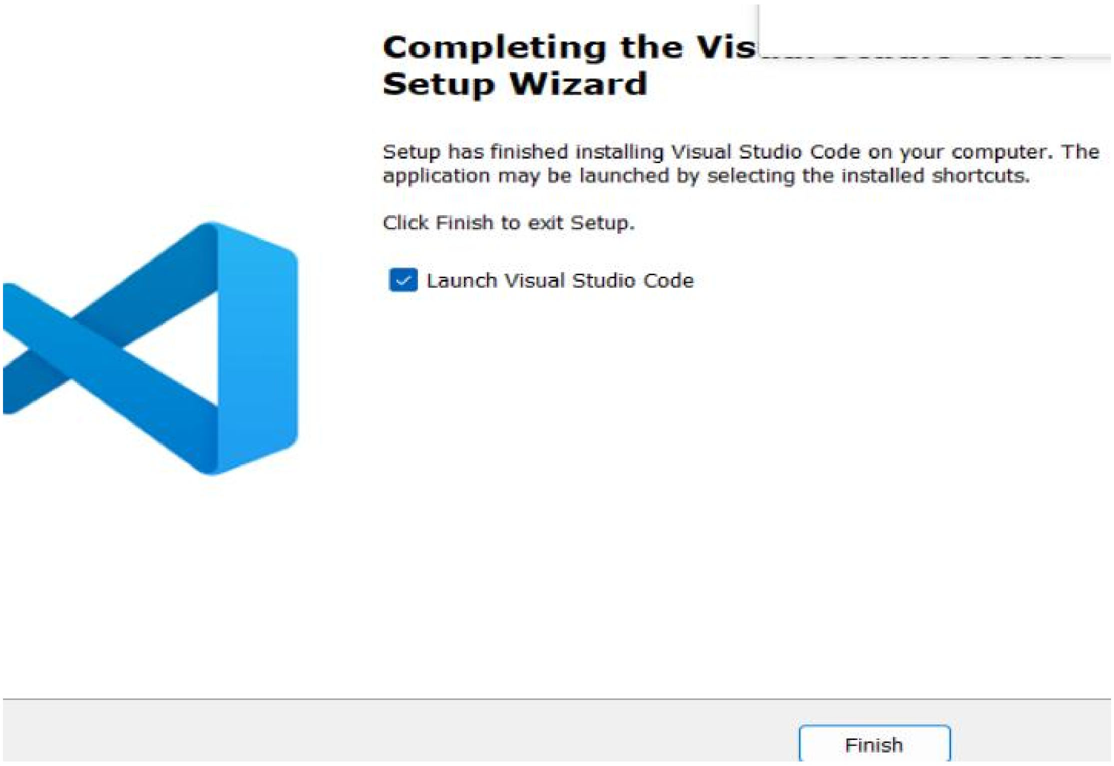
Step 4: Accept the license agreement

1. Select **I accept the agreement**
2. Click **Next**



Step 5: Choose installation location

The default installation location is usually fine.
Click **Next**.

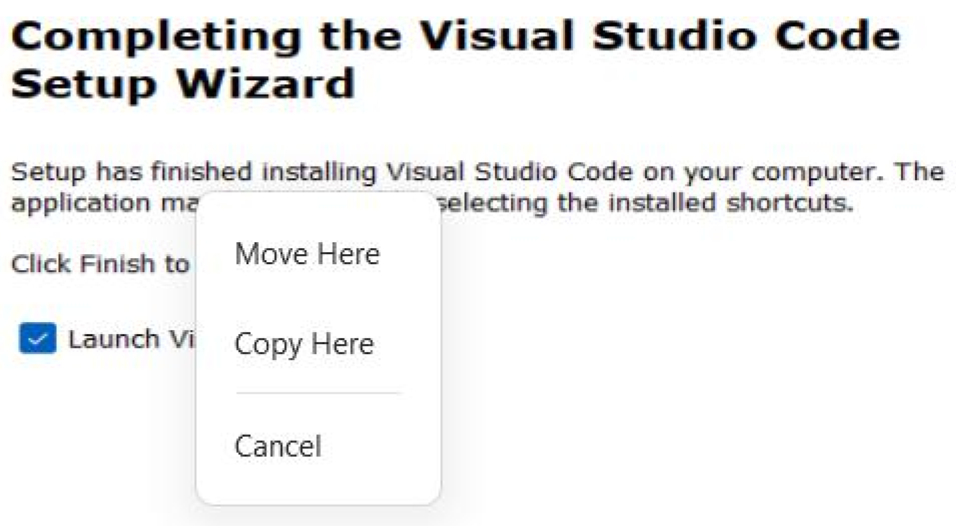


Step 6: Select additional tasks (Important)

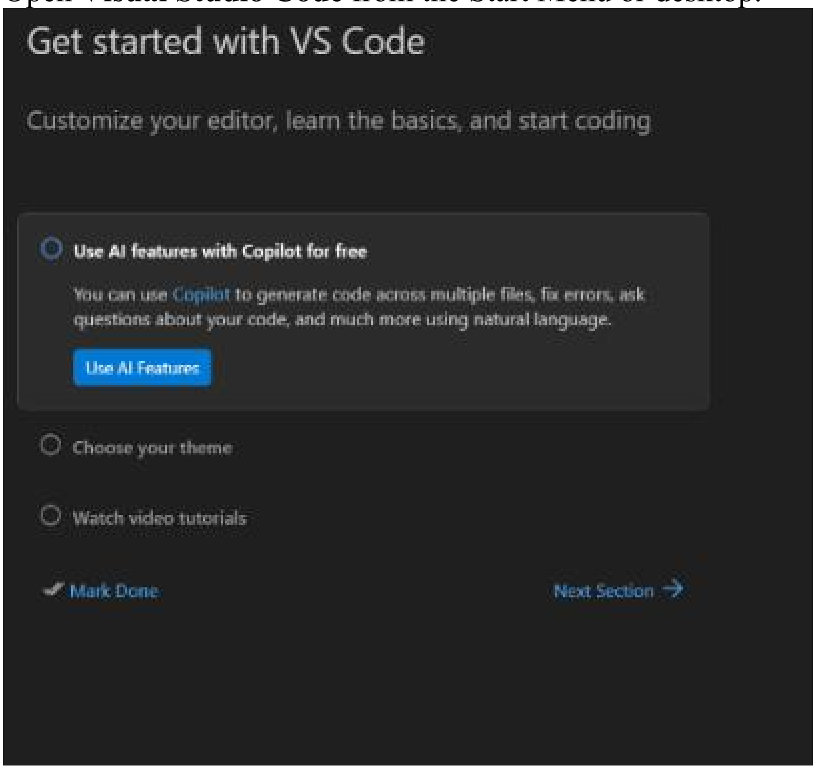
Check the following options:
✔Add **"Open with Code"** action to Windows Explorer (file context menu)
✔Add **"Open with Code"** action to Windows Explorer (directory context menu)
✔**Add to PATH** (Important)
Then click **Next**.

: Install VS Code
Click **Install**.
Wait for the installation process to complete.

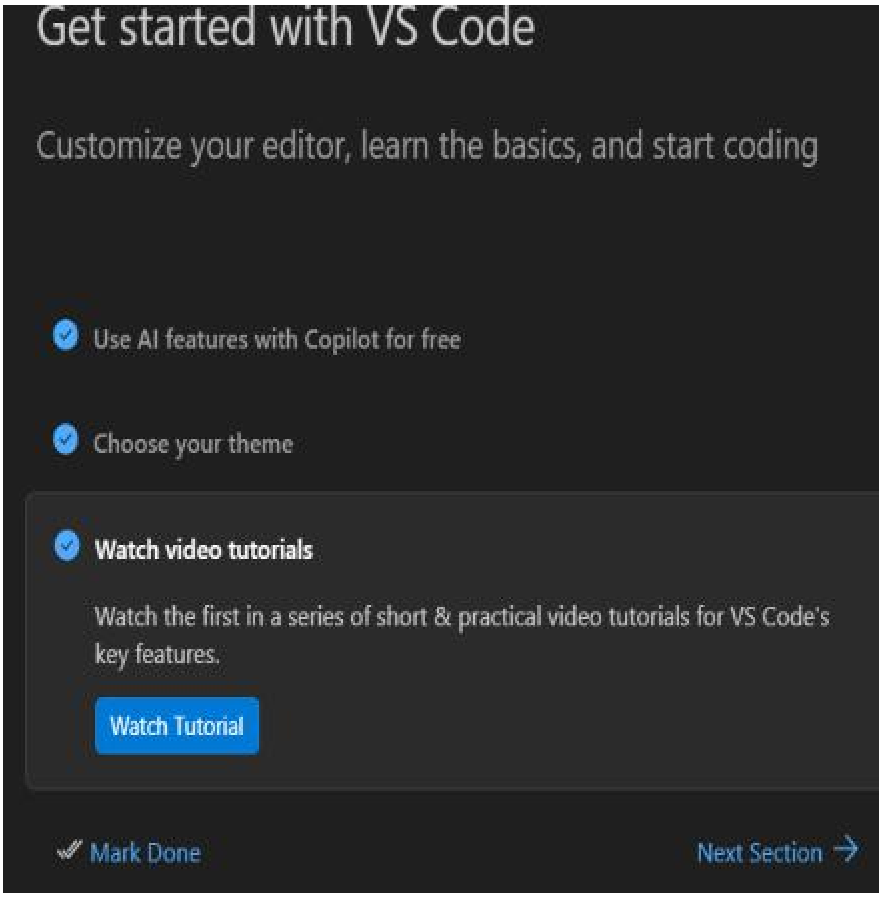
Step 8: Finish installation
After installation is complete:
✔Check **Launch Visual Studio Code**
Click **Finish**.
VS Code will open automatically



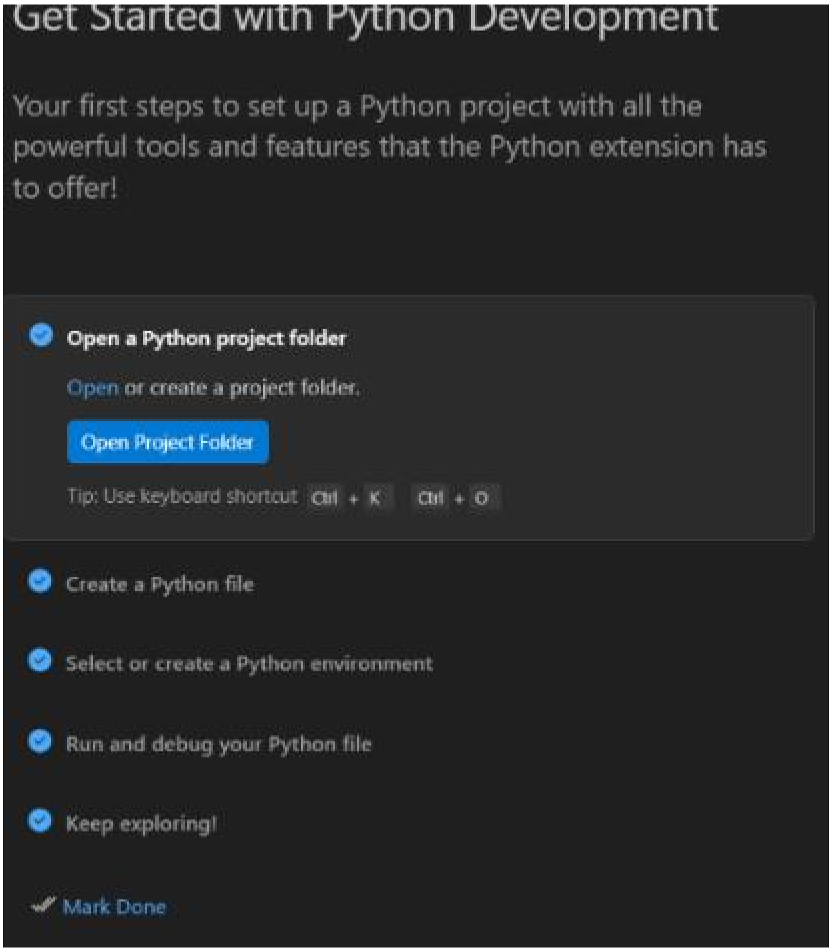
3. Step-by-step procedure to install the Python extension in Visual Studio Code.
Step 1: Open Visual Studio Code
Open **Visual Studio Code** from the Start Menu or desktop.



Step 2: Install the Extension Click the Install button. VS Code will automatically download and install the extension.

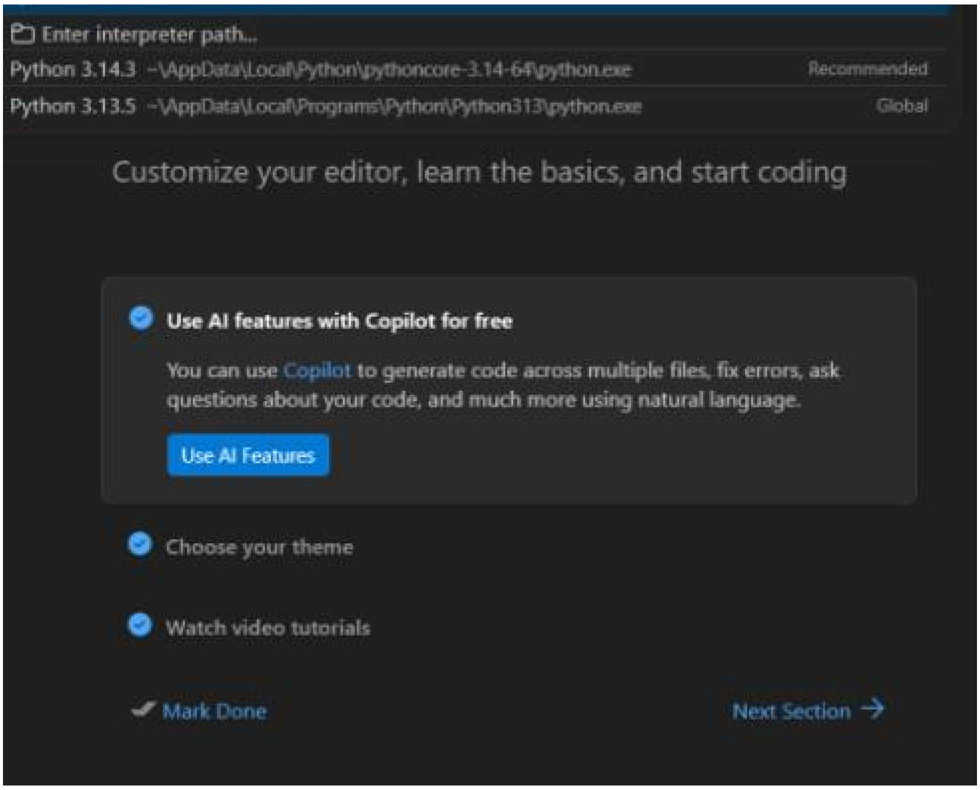


Step 3: Reload VS Code After installation, VS Code may ask you to Reload. Click Reload to activate the extension.

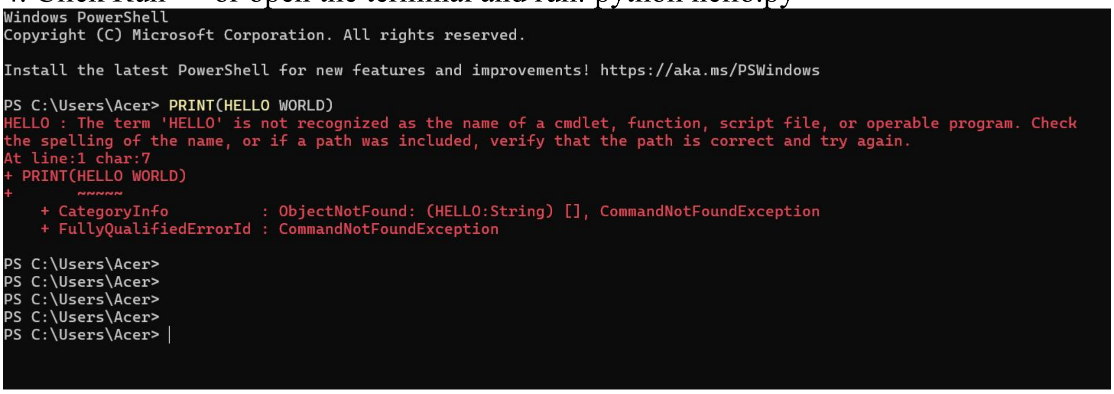


Step 4: Select Python Interpreter Now select the Python interpreter installed on your computer.

1. Press: Ctrl + Shift + P
2. Type: Python: Select Interpreter
3. Choose the installed Python version (example): Python 3.x.x

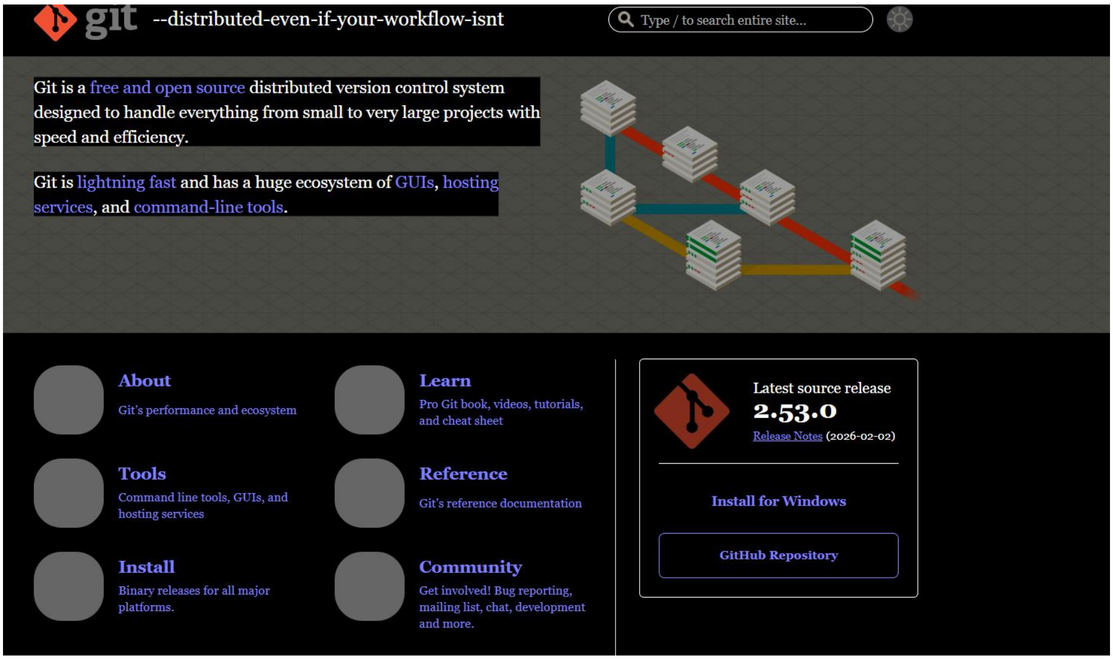


Step 5: Test Python in VS Code
1. Create a new file.
2. Save it as: hello.py
3. Write the following code: print("Hello World")
4. Click Run ▶ or open the terminal and run: python hello.py

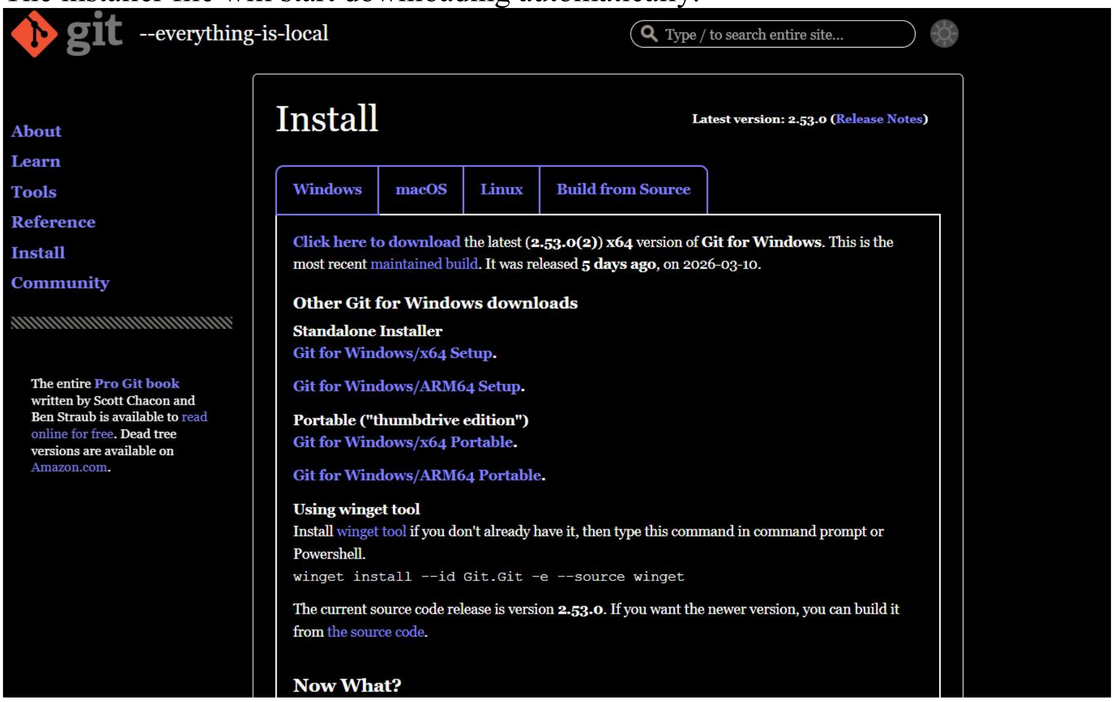


4. Step-by-step procedure to install Git Bash on Windows.

Step 1: Open the Git website Open your web browser and go to the official Git website: https://git-scm.com

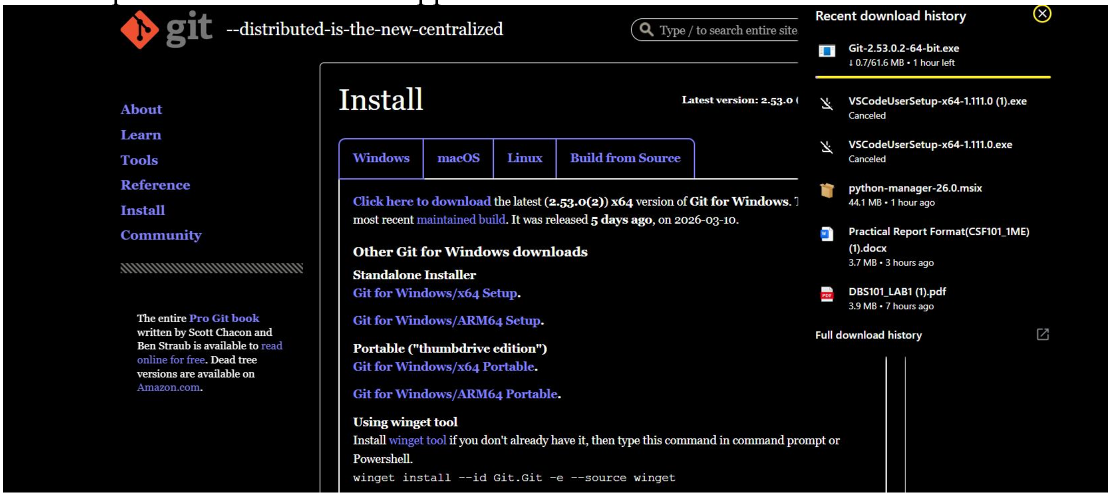


Step 2: Download Git
On the homepage, click the button:
Download for Windows
The installer file will start downloading automatically.



Step 3: Run the installer
After the download is complete, open the installer file:
Git-2.x.x-64-bit.exe

The setup window for **Git** will appear.



Step 4: Complete installation Click Install. Wait for the installation process to finish. And verify installation Open Git Bash and type: git --version If installed correctly.

Conclusion:

What I have learned:

- I learned how to download the different type of programming languages from differents sources.
- To select installation options
- To set up
- Verification
- Installation

Problem faced during the installation:

- Network issues
- Confusion on choosing the best version
- Verification

Reference

Git. (n.d.). https://git-scm.com/

Python releases for Windows. (n.d.). Python.org. https://www.python.org/downloads/windows/

Visual Studio Code - The open source AI code editor. (2021, November 3). https://code.visualstudio.com/